Third Quantum Information Technology and Application Innovation Competition

Mathematical Modeling Innovation Track
Based on Coherent Optical Quantum Computers

Problem B

Innovative Mathematical Modeling of LDPC Decoding in Wireless Communication Networks

Unofficial English translation and editorial compilation
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This document translates the Chinese Problem B statement supplied in the competition materials. Sections 1–6 preserve its background, requirements, example, and reference. Section 7 provides separate editorial clarifications where the source uses ambiguous or inconsistent notation. These clarifications are not additional competition requirements. The Chinese original remains the authoritative source; this document is not an official English release by the organizers.

1 Background

Low-density parity-check (LDPC) codes are efficient error-correcting codes widely used in wireless communication systems. An LDPC code is described by a sparse parity-check matrix. In the notation of the source statement, let N be the number of original information bits and M the number of parity bits. The code rate is the number of input bits divided by the total number of bits:

$$R = \frac{N}{M+N}. \tag{1}$$

The source calls a code a regular (j,k) code if each input bit is connected to k check bits and each check bit is connected to j input bits. This degree description is reproduced here as given; see the notation clarification in Section 7.

Let H denote the LDPC parity-check matrix. Each row of H represents a check-node constraint, and each column specifies the participation of a bit node in those constraints. In Fig. 1, the nodes labeled c_i are check nodes and those labeled b_i are bit nodes. An entry $h_{mn} = 1$ in H indicates an edge between c_m and b_n.

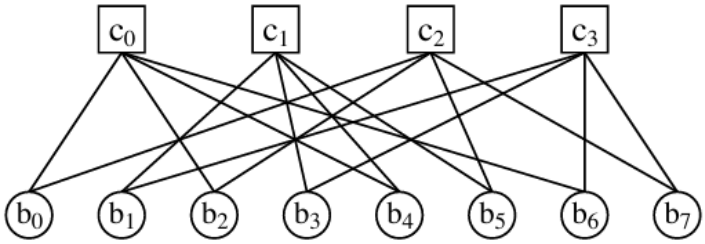


Figure 1: Example Tanner graph for an LDPC code, reproduced from the supplied problem statement.

2 Application Scenarios

Fifth-generation (5G) wireless communication uses two groups of LDPC codes. According to the source statement, one group, BG1, supports a maximum code-block length of 8448 bits and a minimum code rate of 1/3; the other, BG2, supports a maximum code-block length of 3840 bits and a minimum code rate of 1/5.

In 5G NR, the LDPC decoder selects BG1 or BG2 dynamically according to the transport block size (TBS) and code rate R. The task is to design a quantum–classical hybrid decoding architecture that uses quantum computation to accelerate iterative decoding under different signal-to-noise ratios (SNRs), with particular attention to two scenarios:

1. **High-throughput scenario.** BG1-dominated operation, characterized by long code lengths and high code rates: consider a regular $(2, 3)$ code with **180 input bits**.
2. **Low-complexity scenario.** BG2-dominated operation, characterized by short code lengths and low code rates: consider a regular $(3, 6)$ code with **120 input bits**.

3 Core Tasks

Your team is required to design an LDPC decoder based on a quantum algorithm for data transmission in wireless networks. Specifically:

1. **Formulate the decoding problem as QUBO.** Convert LDPC decoding into a quadratic unconstrained binary optimization (QUBO) problem and solve it using a coherent optical quantum computer.
2. **Design and tune the decoder.** Given LDPC parameters, such as code length, code rate, and parity-check matrix, design a quantum decoder capable of handling a specified LDPC code, for example a regular $(3, 6)$ code. Optimize the decoder parameters to improve decoding performance.
3. **Evaluate performance at different SNRs.** Assess the decoder using metrics including bit error rate (BER) and decoding time.

4 Submission and Implementation Requirements

4.1 Decoder Inputs and Outputs

The decoder design must specify both its input data and output data.

Category	Required content
Input: code	LDPC parameters, such as code length, code rate, and parity-check matrix.
Input: channel	Channel parameters, such as SNR and channel state information.
Input: signal	Received signal: the LDPC-encoded signal affected by noise.
Output: bits	Decoded bit sequence.
Output: metrics	Decoding performance indicators, such as BER and decoding time.

4.2 Variable Code Lengths and Larger Problems

The decoder must support variable code lengths. Participants may choose lengths N larger than those specified in the two scenarios for regular $(2, 3)$ and $(3, 6)$ codes, and are encouraged to address large-scale problems.

Increasing code length can rapidly increase computational requirements. Explore how the model can be changed to solve larger problems while keeping the required computational resources unchanged or only slightly increased.

4.3 Data Generation and Solver

No dataset is provided for this problem. Each team must generate its own data according to the requirements above. The code length should be selected with reference to the number of bits supported by the platform.

Use the **simulated solver in the Kaiwu SDK** to solve the problem. The SDK download location given in the source is:

`https://platform.qboson.com/sdkDownload`

4.4 Required Deliverables

Provide the complete project code and a project report. The report must describe the model construction in detail and include the solution time.

5 Illustrative Decoder Example

The source provides the following example dataset for testing and validating the decoder. Its numerical dimensions and wording are preserved here; their interpretation is discussed separately in Section 7.

5.1 Example Input

- **Bit sequence to be decoded:** 120 bits of binary data.
- **Code length:** 120 bits.
- **Code rate:** 1/2.
- **Parity-check matrix:** a 120×240 matrix generated according to a regular $(3, 6)$ LDPC code.
- **SNR:** perform decoding separately at 9 dB, 15 dB, and 35 dB.

5.2 Example Output

- **Decoded bit sequence:** `010101...` (120 bits).
- **Decoding performance indicators:** BER and decoding time.

6 Reference Supplied in the Problem

S. Kasi and K. Jamieson, "Towards quantum belief propagation for LDPC decoding in wireless networks," in *Proceedings of the 26th Annual International Conference on Mobile Computing and Networking (MobiCom)*, 2020, pp. 1–14.

Verified bibliographic identifier: doi: 10.1145/3372224.3419207.

7 Editorial Clarifications

Status of this section. These notes explain ambiguous notation. They do not amend the competition rules or represent an official clarification issued by the organizers.

7.1 Information Length and Codeword Length

The source first uses N for the number of information bits, but later uses N when discussing code length. Standard coding notation commonly distinguishes information length K from codeword length n. If $H \in \mathbb{F}_2^{m\times n}$, then

$$K = n - \operatorname{rank}_{\mathbb{F}_2}(H), \qquad R = K/n. \tag{2}$$

Report whether the scenario's 180 or 120 bits mean information bits or encoded bits, together with the matrix dimensions and rank. A check equation need not correspond to an independent, explicitly transmitted parity bit.

7.2 Degree Convention for Regular Codes

The background paragraph describes (j, k) using bit-node degree k and check-node degree j. Many LDPC references and the supplied project code instead use column degree j and row degree k. Under the latter convention,

$$nj = mk, \qquad R_{\text{design}} = 1 - \frac{m}{n} = 1 - \frac{j}{k}. \tag{3}$$

This gives design rates $1/3$ for $(2, 3)$ and $1/2$ for $(3, 6)$. State the chosen degree convention explicitly; the labels alone do not determine actual rate or computational complexity.

7.3 Dimensions of the Example

The example simultaneously states a 120-bit code length, a rate of $1/2$, and a parity-check matrix of size 120×240. With the usual row-check/column-bit convention, that matrix has **240 codeword bits**. If it has full row rank, it defines a code with **120 information bits** and rate $1/2$.

Thus $K = 120$, $n = 240$, $m = 120$ is one consistent interpretation. Alternatively, $n = 120$ at rate $1/2$ means $K = 60$, typically with a full-row-rank 60×120 matrix. The source does not choose between these interpretations; its original example is preserved above.

7.4 BG1/BG2 Terminology

The source's 8448-bit and 3840-bit figures are associated with maximum code-block sizes in the NR coding construction; they should not be read as full lifted mother-codeword lengths. Standard BG1 and BG2 have base-matrix sizes 46×68 and 42×52. Before the specified puncturing and rate matching, lifting by Z gives $68Z$ and $52Z$ columns, respectively.

The regular-code examples do not specify 3GPP shift tables, lifting factors, filler bits, or rate matching. Distinguish these modeling scenarios from the full 5G NR coding chain. See 3GPP TS 38.212, *NR; Multiplexing and channel coding*: `https://www.3gpp.org/dynareport/38212.htm`.

7.5 Simulation, Platform Budget, and Evaluation

The source explicitly requires an SDK *simulated* solver within a coherent-optical-computing modeling workflow. Identify simulation and physical hardware runs separately. No universal numerical hardware limit is specified. Count data variables, QUBO auxiliaries, and backend conversion overhead against the selected platform budget.

Document the SNR convention (E_s/N_0, E_b/N_0, or another definition), channel model, modulation, sample count, and stopping rule; these are not fixed by the source. Compute BER against the known transmitted sequence. State which processing stages and solver backend are included in decoding time.

Source and editorial record. Translated from the Problem B Word document in the supplied competition archive, retaining the original Tanner graph. The reference DOI and Section 7 are editorial additions. This document contains the problem statement, not a competition solution.

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